

All spectral frequencies of neural activity reveal semantic representation in the human anterior ventral temporal cortex

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Abstract:

Intracranial electrophysiology offers a unique insight into the nature of information representation in the brain – it can be used to disentangle information encoded in gamma and high gamma frequencies from information encoded in lower frequencies. We used regularised logistic regression to decode animacy from time-frequency power and phase extracted from electrocorticography (ECoG) grid electrode data recorded on the surface of human vATL. Power in gamma (30 – 60 Hz) and high gamma (60 – 200 Hz) produced reliable decoding, indicating that semantic information is indeed expressed by local populations in vATL. However, power from a wide range of frequencies (4 – 200 Hz) produced significantly higher decoding accuracy and also exhibited the same rapidly-changing dynamic code previously observed when decoding voltage. These findings support the theory that semantic information is encoded by a local vATL “hub” that interacts with distributed cortical “spokes”.

1. Introduction

Semantic cognition is our ability to understand and respond in a meaningful way to objects that we see, language that we hear, events that we experience, and any other stimuli in our environment. The ventral anterior temporal lobes (vATLs) play a critical role in such abilities (Lambon Ralph et al., 2017; Patterson et al., 2007; Rogers et al., 2004), but it remains unclear *how* neural activity in these regions encodes semantic information. Electrocorticography (ECoG) grid electrode data are a rare and informative resource, sensitive to activity at a wide range of frequencies including very high-frequency activity thought to indicate local neuronal activity (Başar-Eroglu et al., 1996; Bressler, 1990; Merker, 2013; Nir et al., 2007). Although prior work has shown that surface voltages express information about stimulus animacy (“is it an animal or not?”; Y. Chen et al., 2016; Rogers et al., 2021), voltages are a conflation of high- and lower-frequency activity; therefore, apparent representation of semantic information by the vATL may actually reflect incoming signals from more remote areas. We address these questions by applying regularised logistic regression — a multivariate decoding procedure that makes minimal assumptions about the nature of neural representations (Cox & Rogers, 2021; Frisby et al., 2023) — to time-frequency power and phase extracted from a large sample of ECoG data ($n = 18$).

Extensive prior work suggests that semantic cognition is underpinned by multiple “spokes” that each encode modality-specific information about the environment, together with a “hub” that synthesises information across modalities and temporal instances and thereby enables generalisation of knowledge across items and contexts (Lambon Ralph et al., 2010, 2017; Patterson et al., 2007; Rogers et al., 2004). In particular, neuropsychological studies have established that patients with bilateral atrophy of the vATLs exhibit impairments on semantic tasks, irrespective of the input and/or output modality, while other abilities are spared (Bozeat et al., 2000; Hodges et al., 1992; Hodges & Patterson, 2007). This supports the conclusion that the vATLs (bilaterally) are the location of the semantic hub. Additionally, the “hub-and-spoke” model is supported by convergent evidence from positron emission tomography (PET; Devlin et al., 2000), distortion-corrected functional magnetic resonance imaging (fMRI; Binney et al., 2010; Halai et al., 2014; Visser et al., 2010), magnetoencephalography (MEG; Mollo et al., 2017), neuro-computational

simulations (L. Chen et al., 2017; Rogers et al., 2004, 2021), transcranial magnetic stimulation (TMS; Pobric et al., 2007, 2010a, 2010b), invasive cortical stimulation mapping (Lüders et al., 1991; Matoba et al., 2024), and ECoG (Halgren et al., 2006; Nobre & McCarthy, 1995; Sato et al., 2021; Shimotake et al., 2015).

Multivariate decoding approaches have only recently been applied to ECoG data in an attempt to characterise, not simply where and when task-related changes in neuronal activity occur, but what information those changes represent. Most of these analyses have focused on voltage (Y. Chen et al., 2016; Cox et al., 2024; Liu et al., 2009; Nagata et al., 2022; Rogers et al., 2021; cf. Clarke, 2020; Rupp et al., 2017; Wang et al., 2011). However, voltage reflects a blend of activity at different frequencies and, while gamma (> 30 Hz) and high gamma (> ~60 Hz) activity is thought to reflect local neuronal activity (Başar-Eroglu et al., 1996; Bressler, 1990; Merker, 2013; Nir et al., 2007), activity at lower frequencies may reflect transmission of information over varying cortical distances (Canolty & Knight, 2010; Kopell et al., 2000; Lachaux et al., 2012), which may be important for assembling the spatially-distributed representations posited by the hub-and-spoke model.

The first aim of this paper was to clarify the role of time-frequency power and phase in semantic representation in the vATL and to relate this to voltage findings (Y. Chen et al., 2016; Cox et al., 2024; Liu et al., 2009; Nagata et al., 2022; Rogers et al., 2021) and to other sources of evidence (Lambon Ralph et al., 2017). The dataset consisted of ECoG data collected from a large cohort of patients ($n = 18$) while they named line drawings of animate and inanimate objects. ECoG offers excellent spatial and temporal resolution and the use of grid electrodes on the cortical surface offers much greater coverage than can be obtained with cortical depth electrodes — important for detecting semantic representations that may be distributed across space (Merker, 2013; Rogers et al., 2021). Crucially for this study, intracranial methods are necessary for revealing high gamma activity that is blocked by the skull and scalp when using noninvasive methods such as EEG or MEG (Lachaux et al., 2012; Parvizi & Kastner, 2018).

We used regularised logistic regression to decode information from time-frequency power and phase — a method that, unlike decoding methods that have previously been applied to ECoG time-frequency data (Clarke, 2020; Rupp et al., 2017; Wang et al., 2011) makes very few assumptions about the properties of the underlying code (Frisby et al.,

2023). Regularised regression can detect representations whether individual populations or frequencies encode pieces of information separately, or whether the relationship between information and neural activity can be understood only by considering multiple populations and/or frequencies simultaneously (Frisby et al., 2023). Since it relies on parameter fitting rather than simple correlations, it is resistant to false positives and false negatives associated with purely correlational approaches (Cox et al., 2024).

We tested whether animacy (a) can be decoded from time-frequency power and/or phase; (b) if so, from which frequency ranges and at what timepoints; and (c) how these results relate to decoding of voltage. These analyses therefore allowed us to determine whether time-frequency- and voltage-based decoding offer similar or different conclusions about the timecourse of semantic representation in the vATL. Decoding accuracy in gamma and high gamma ranges (independently of other ranges) allowed us to determine whether semantic information is encoded by local neural activity within the vATL.

The second aim of this paper was to assess whether the dynamic and nonlinear changes to semantic representation observed in voltage decoding (Rogers et al., 2021) also arise within or across frequency ranges. Dynamic change in representation can be measured via the *temporal generalisation* of a decoding model (Carlson et al., 2013; Cichy et al., 2014; Contini et al., 2017; King & Dehaene, 2014; Rogers et al., 2021). The model is fitted to neural data at a given timepoint, then is assessed at other timepoints. If a classifier shows above-chance hold-out accuracy when fitted to timepoint A, and also shows above-chance accuracy when used to decode at timepoint B, we can conclude that information is encoded similarly at both timepoints. Conversely, if a classifier trained and succeeding in timepoint A then fails when tested on timepoint B, and if the reverse pattern also holds (a model trained on B succeeds on B but fails on A), this suggests that the target information is encoded by different neural patterns at different timepoints. That is, such a pattern suggests dynamic change in neural representation of the target information.

When applied to ECoG voltage data, this technique revealed four interesting characteristics of the semantic code in the vATL (Rogers et al., 2021):

1. Classifiers fitted and tested at overlapping 50 ms windows showed reliable decoding from about 100ms post stimulus onset through the remainder of the processing

window (1650ms), suggesting that animacy is *constantly decodable* from the moment stimulus-driven activity reaches the vATL until the response.

2. Classifiers trained at a given window generalised to temporal neighbours (both preceding and succeeding), but not to temporally remote windows; that is, voltage-based neural decoding showed *local temporal generalisation*, suggesting a dynamically-changing neural code for animacy.

3. The width of the temporal generalisation window for a given classifier was quite narrow early in stimulus processing but widened over time, indicating that dynamic change to the voltage-based code is initially quite rapid but slows with further processing.

4. The direction of the neural code — whether an increase in voltage at a particular electrode signified increased or decreased probability that the stimulus is animate — changed over the course of processing, so that voltage deflections might signify different things at different timepoints at the same electrode.

Together these properties suggest that, while information about animacy is expressed in voltage throughout stimulus processing, the precise way that this information is encoded changes rapidly and nonlinearly, with change slowing as the system settles into a steady state. Moreover, these patterns mirrored those observed in a recurrent deep neural network model of semantic processing, suggesting that such models may offer a useful way of thinking about representational dynamics in real neural systems (Rogers et al., 2021).

The second goal of this paper was to assess whether these properties (constant decodability, local temporal generalisation, widening generalisation window, and change in code direction) are observed when decoding time-frequency power or phase information, across ranges or within each range considered independently. Answers to these questions could clarify whether these dynamic characteristics of representation arise from rapid local activity (i.e., driven by gamma/high gamma) and/or from more remote input (driven by lower frequencies).

2. Methods

2.1. Patients

19 patients participated in the study (labelled 01-22 to enable comparison with previous work (Y. Chen et al., 2016; Cox et al., 2024; Rogers et al., 2021; Shimotake et al., 2015). Of these 19 patients, one was later excluded because too many trials were contaminated with artefacts (see Preprocessing – ECoG), leaving 18 patients for decoding. For one further patient (Patient 20), no exact MNI coordinates were available for their electrodes. This patient was included in the analyses, but excluded from visualisation (see Experimental Questions). All patients were native speakers of Japanese. Information about patients' age, sex, handedness, and clinical presentation is summarised in Table 1.

Each patient was implanted with subdural electrodes arranged in grids or strips for presurgical monitoring (mean 88 electrodes, range 56 – 108 electrodes per patient). 15 patients had electrodes implanted in the left hemisphere, of which 10 – 42 electrodes (mean 25 electrodes) were within grids or strips that covered the vATL (where grids or strips included both electrodes that covered ventral anterior temporal cortex and electrodes that covered lateral or posterior temporal cortex, the whole grid or strip was included in the analysis). In the remaining 3 patients, with electrodes implanted in the right hemisphere, 6 – 28 electrodes (mean 20 electrodes) were within grids or strips that covered the vATL. Grids in which no electrodes covered the vATL were excluded from analysis. Electrodes that were included in the analysis are shown in Figure 1. The electrodes were platinum, with a recording diameter of 2.3 mm and an inter-electrode distance of 1 cm (ADTECH, WI). Four patients were also implanted with depth electrodes (sEEG), but these were not included in the analysis.

All patients gave written informed consent and the study was approved by the ethics committee of the Kyoto University Graduate School of Medicine (#C533).

	Patient 01	Patient 02	Patient 03	Patient 04
Age, sex, handedness	22, M, R	29, M, R&L	17, F, R	38, F, R
WAIS-R/WAIS-III* (VIQ, PIQ, TIQ)	70, 78, 69	72, 78, 72	67, 76, 69	84, 97, 89
WMS-R (verbal, visual, general, attention, delayed recall)	99, 64, 87, 91, 82	99, 92, 97, 87, 83	51, <50, <50, 81, 56	75, 111, 83, 62, 53
WAB	95.6	96	97.2	98.5
WADA	Left	Bilateral	Left	Left
Age of seizure onset	16	10	12	29
Seizure type	FAS→FIAS, FBTCS	FAS→FIAS	FAS→FIAS	FAS→FIAS
Ictal ECoG onset	aMTG	PHG	PHG	PHG

MRI	L basal frontal cortical dysplasia, L anterior temporal arachnoid cyst	L posterior temporal cortical atrophy	L temporal tip arachnoid cyst	L HS/HA
Pathology	FCD type I	FCD type IIIa	Palmini FCD type IB	FCD type IIa
	Patient 05	Patient 06	Patient 07	Patient 08
Age, sex, handedness	55, M, R	34, M, L	41, F, R	27, F, R
WAIS-R/WAIS-III* (VIQ, PIQ, TIQ)	105, 99, 103	55, **, 44	72, 83, 75	106, **, 105
WMS-R (verbal, visual, general, attention, delayed recall)	71, 117, 84, 109, 72	52, <50, <50, 55, <50	83, 111, 89, 94, 82	112, 114, 114, 81, 100
WAB	98	88	97.3	99.6
WADA	Left	Left	Right	Left
Age of seizure onset	55	12	19	16
Seizure type	FIAS (once)	FAS→FIAS	FAS→FIAS	FAS→FIAS
Ictal ECoG onset	None	R parietal lobe/pMTG	PHG	ventral anterior temporal
MRI	L medial temporal lobe low-grade glioma	R parietal cerebral atrophy & contusion, R hippocampal sclerosis/atrophy	L HS/HA, L parieto-occipital perinatal infarction	R medial temporal cyst
Pathology	Diffuse astrocytoma	Post-traumatic change(parietal)/scar(temporal)/ HS	FCD type I	FCD type I
	Patient 09	Patient 10	Patient 11	Patient 12
Age, sex, handedness	51, M, R	38, F, R	29, F, R	40, M, L
WAIS-R/WAIS-III* (VIQ, PIQ, TIQ)	73, 97, 83	109, 115, 112	62, 80, 67	93, 105, 98
WMS-R (verbal, visual, general, attention, delayed recall)	80, 101, 85, 91, 91	71, 79, 70, 90, 58	64, 94, 68, 79, 79	74, 94, 77, 110, 96
WAB	89.6	96.9	95.8	99
WADA	Left	Left	Left	Right
Age of seizure onset	43	28	12	6
Seizure type	FIAS	FAS→FIAS	FAS→FIAS	FAS→FIAS
Ictal ECoG onset	mITG	SMG	PHG	PHG
MRI	L temporal cavernoma	L parietal operculum tumour	L HS	R HS/HA
Pathology	Arteriovenous malformation	Oligoastrocytoma	non-neoplastic brain tissue	FCD type IIIa
	Patient 13	Patient 14	Patient 15	Patient 20
Age, sex, handedness	22, M, R	42, M, R	35, M, R	23, F, R
WAIS-R/WAIS-III* (VIQ, PIQ, TIQ)	86, 79, 81	96, 84, 90	82, 86, 82	67, 82, 71
WMS-R (verbal, visual, general, attention, delayed recall)	55, 79, 53, 90, 54	73, 85, 73, 103, 76	75, 89, 75, 92, 81	78, 119, 86, 82, 64
WAB	97.4	99.9	99.2	94.2
WADA	Left	Left	Left	Bilateral
Age of seizure onset	14	27	20	15
Seizure type	FAS→FIAS	FAS→FIAS	FIAS	FIAS
Ictal ECoG onset	PHG	PHG	PHG	IPL/SMG
MRI	L HS/HA	L HS	L HS/HA	No apparent lesion
Pathology	FCD type IIIa	FCD type IIIa	FCD type IIIa	FCD type I
	Patient 21	Patient 22		

Age, sex, handedness	40, M, R	28, F, R
WAIS-R/WAIS-III* (VIQ, PIQ, TIQ)	86, 97, 90	80, 69, 72
WMS-R (verbal, visual, general, attention, delayed recall)	111, 113, 113, 87, 109	77, 89, 77, 121, 73
WAB	92.2	100
WADA	Left	Left
Age of seizure onset	30	12
Seizure type	FIAS	FIAS
Ictal ECoG onset	aMTG	PHG
MRI	No apparent lesion	L HS/HA
Pathology	FCD type I	FCD type IIIa

Table 1: Patient characteristics. Abbreviations: WAIS-R – Wechsler Adult Intelligence Scale (1991), WAIS-III - Wechsler Adult Intelligence Scale (1997), VIQ – Verbal IQ, PIQ – Performance IQ, TIQ – full-scale IQ, WMS-R – Wechsler Memory Scale (1987), WAB – Western Aphasia Battery, FAS – focal aware seizure, FIAS – focal impaired awareness seizure, FBTCS – focal to bilateral tonic-clonic seizure, aMTG – anterior middle temporal gyrus, PHG – parahippocampal gyrus, pMTG – posterior middle temporal gyrus, mITG – medial inferior temporal gyrus, SMG - supramarginal gyrus, ITG – inferior temporal gyrus, IPL – intraparietal lobule, HS – hippocampal sclerosis, HA – hippocampal atrophy, FCD – focal cortical dysplasia. * - The WAIS-R was used to test patients 01-06 and the WAIS-III was used to test other patients. ** - missing score.

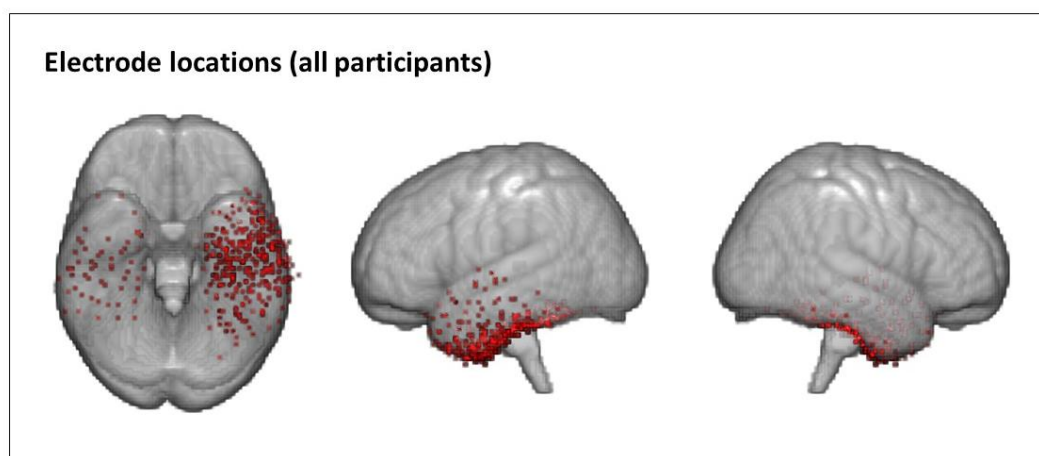


Figure 1: Electrode locations for all patients. Electrodes are overlaid on an MNI template (MNI152Nlin2009cAsym).

2.2. Stimuli, task, and acquisition

Stimuli were the same 100 line drawings used previously (Y. Chen et al., 2016; Rogers et al., 2021; Shimotake et al., 2015) – 50 animals and 50 nonliving items including buildings, tools, musical instruments and other household objects (Morrison et al., 1997); https://github.com/slfrisby/ECoG_LASSO/tree/main/data_info/stimuli/. There were no

significant differences between the categories with respect to age of acquisition, visual complexity, concreteness, familiarity, word frequency, and name agreement (Barry et al., 1997; Rogers et al., 2021). Particular care was taken to ensure that animacy could not be predicted based on word frequency or on non-semantic visual structure. Logistic regression classifiers identical to those used in the main analysis (see section 2.3.3.1) were trained to discriminate animate from inanimate stimuli using word frequencies as input features (Morrison et al., 1997). In previous work, the same approach was taken using vectors describing visual structure, obtained with Chamfer matching (Rogers et al., 2021). In neither case was the performance of these classifiers significantly different from chance, confirming that any differences between stimuli in word frequency or visual structure would not be able to account for a subsequent positive decoding result. MATLAB r2010a was used to display stimuli on a PC screen.

There were four runs per patient, collected in a single session. Within each run, each stimulus was presented once in a random order. Stimuli were presented for 5 seconds each, with no break between stimuli.

Patients were instructed to name each picture as quickly and accurately as possible. Data for nine patients were recorded at 2000 Hz (with a low-pass filter of 600 Hz) and data for ten patients were recorded at 1000 Hz (with a low-pass filter of 300 Hz). Time of naming onset was measured. Responses and eye fixation were monitored via video.

2.3. Data analysis

2.3.1. Preprocessing – structural MRI

A clinical MPAGE T₁-weighted anatomical scan was acquired before and after electrode implantation. The location of each electrode was identified on each 2D slice of the post-surgical scan. *fnirt* in FSL (<https://fsl.fmrib.ox.ac.uk/fsl/fslwiki/>; Jenkinson et al., 2012; Smith et al., 2004) was used to coregister the electrode positions to the pre-surgical scan and then to MNI space (MNI152NLin2009cAsym). The position of each electrode was then manually adjusted to the surface.

2.3.2. Preprocessing – ECoG

Preprocessing was performed using in-house MATLAB (r2023b) code, composed of functions from EEGLAB (2023.1; <https://eeglab.org/>; Delorme & Makeig, 2004) and available at https://github.com/slfrisby/ECOG_LASSO/. First, the CleanLine EEGLab plugin (v2.0; <https://github.com/scn/cleanline/>; Delorme, 2023; Mitra & Bokil, 2007) was used to remove line noise at 60 Hz and the harmonics 120 and 180 Hz, without scanning for exact line noise frequencies and with a sliding window step size of 2 (all other parameters were set as defaults). CleanLine uses a multitaper fast Fourier transform to identify and remove sinusoidal noise and, though it removes noise less completely than traditional notch-filtering, is less disruptive to the overall frequency structure of the data. Data were then filtered using *eegfilt* (EEGLAB). The low cutoff was set to 0.5 Hz (to remove slow drifts; Delorme, 2023) and the high cutoff set to 300 Hz (for consistency across patients - some data were recorded at 1000 Hz using equipment that imposed a low-pass filter at 300 Hz). Channels that lay below the seizure onset zone (identified in the clinical examinations) or with poor contact (identified by visual inspection) were removed. Data were epoched between -1000 and 3000 ms relative to stimulus onset and baseline-corrected using the mean response across trials between -200 and -1 ms. Data from the nine patients recorded at 2000 Hz was then down-sampled to match the ten patients recorded at 1000 Hz by boxcar averaging pairs of neighbouring timepoints. A common average reference was applied using *reref* (EEGLAB).

Addressing artefacts in the data was a five-step process. First, the mean voltage for each channel was calculated over timepoints and trials. Any trial that contained a value more extreme than 10 standard deviations from the mean was rejected. Second, all trials were inspected individually and any that appeared to contain obvious interictal epileptiform activity, muscle activity, “electrode pop” or other artefacts were rejected. Since repeated presentations of the same stimulus were to be averaged it was important to check whether there was at least one good trial for each stimulus. One patient lacked any good trials for 18 stimuli (7 animate, 11 inanimate) and so was excluded from further analysis (and does not appear in Table 1). Another lacked any good trials for 3 stimuli (1 animate, 2 inanimate) – because of the small and relatively balanced number of missing stimuli, we included this patient. For the next steps an independent component analysis (ICA) was conducted using *runica* (EEGLAB) extracting n components, where n was 75% of the total number of good

electrodes (Clarke, 2020). For the third artefact rejection step, the SASICA EEGLAB plugin (<https://github.com/dnacombo/SASICA/>; Chaumon et al., 2015) was used to identify ICA components with weak autocorrelation, which is characteristic of muscle activity. Components identified by SASICA were inspected and, in order to minimise rejection of signal-carrying data, rejected if the channel appeared to contain only autocorrelated noise and no neural activity (i.e., only very-high-frequency activity that remained constant in amplitude over time - no low-frequency activity or amplitude variation). Since no components met this criterion, no components were rejected based on this assessment. Fourth, SASICA was used to highlight components with focal trial activity (i.e., activity in only a few trials, characteristic of muscle activity), “electrode pop” or other artefacts. Rather than rejecting the component across all trials, the atypical trials were identified via visual inspection and only those trials were removed. The ICA was then re-run and the process repeated until no components contained significant focal trial activity. Fifth, the possibility of microsaccade artefacts in the data was assessed (Clarke, 2020; Jerbi et al., 2009; Kovach et al., 2011; Yuval-Greenberg et al., 2008). The independent component activations were filtered for activity in the frequency range associated with microsaccades (20 – 190 Hz). The activations were then convolved with a saccade-related potential template and the number of microsaccades per second was calculated (Craddock et al., 2016). Since every component contained very few microsaccades (< 0.000012 microsaccades per second) and no component obviously contained more microsaccades than any other, no components were rejected based on this assessment.

Time-frequency power and phase were extracted using complex Morlet wavelet convolution (Bertrand et al., 1994), implemented with *timefreq* (EEGLAB). Power values were extracted from every trial between 0 and 1650 ms in 10 ms time steps and between 4 and 200 Hz in 60 logarithmically-spaced frequency steps. A 5-cycle wavelet was used at 4 Hz, increasing to a 15-cycle wavelet at 200 Hz (Clarke, 2020). Power values were then averaged across repeated presentations of the same stimulus and any missing trials were interpolated with the median power value across all trials. Decibel normalisation was performed using the mean power across trials between -300 and -100 ms (calculated separately for each electrode and frequency) – a gap before stimulus onset was used to mitigate the effect of temporal leakage of trial activity into the baseline during wavelet convolution. Since phase

values cannot be averaged (Cohen, 2014), preprocessed trials were averaged across repeated presentations of the same stimulus before phase values were extracted using the same time and frequency steps used for the extraction of power. As a comparison, preprocessed voltage was averaged over repeated presentations of the same stimulus.

The impact of preprocessing decisions on decoding results is explored in Supplementary Materials S2. In summary, decoding of semantic structure (animacy) with spectral frequency features remains unchanged after each step in this extensive and careful cleaning.

2.3.3. Multivariate classification

2.3.3.1. Decoding approach

Logistic regression classifiers were trained to discriminate animate from inanimate stimuli using *glmnet* in R (Friedman et al., 2010). For power and phase, frequency feature vectors were created for each stimulus at each timepoint (0 ms, 10 ms, 20 ms, ...) by concatenating power or phase values for all electrodes for all frequencies in a range of interest. Voltage feature vectors were created for each stimulus at each timepoint by concatenating voltage values for all electrodes in a 50 ms window centred on the timepoint of interest. The voltage feature vectors therefore reflected activity around the timepoint of interest – but note that so did frequency feature vectors, since Morlet wavelet convolution takes activity at neighbouring timepoints into account. These vectors were provided as input to the classifiers. Each classifier was trained on vectors generated at a single timepoint for a single patient.

The classifiers used LASSO (L1) regularisation (Tibshirani, 1996), which applies a penalty that scales with the sum of the absolute values of the coefficients and thus produces solutions in which many features receive coefficients of zero. This approach was selected because it can be used to assess whether the same electrodes or frequencies are used to represent information at different timepoints. If the information used by a classifier trained at timepoint A is present at timepoint B, the classifier will perform well at timepoint B; other units, that may be in different states at the two timepoints, will receive coefficients of zero and therefore will not affect classifier performance (Rogers et al., 2021).

Classifier accuracy was assessed using 10-fold cross-validation. In each outer loop, 10 out of 100 stimuli (5 animate, 5 inanimate) were held out. The remaining 90 stimuli were used to search a range of 100 values (0.2 – 0.002, logarithmically spaced) for the regularisation parameter that resulted in the smallest mean squared error. This process was implemented using *cv.glmnet* (*glmnet*) with parallel model fitting implemented with *foreach* (<https://cran.r-project.org/web/packages/foreach/index.html>; all other parameters were set as defaults). A model with the best regularisation parameters was tested on the 10 stimuli in the outer loop hold-out set. The same model was tested on the same 10 stimuli at all other timepoints. The process was then repeated 10 times with different final hold-out sets. This process yielded both a main timecourse of decoding accuracy (i.e., mean accuracy, over folds, of classifiers tested on the timepoint at which they were trained) and a generalisation matrix in which the y-coordinate of a cell is the time at which the classifier was trained and the x-coordinate is the time at which the classifier was tested. Finally, a single classifier was trained at every timepoint using all 100 stimuli for training. The accuracy of these classifiers was not evaluated – they were used only for inspection of coefficients.

To assess group-level performance, the timecourses of decoding accuracy and the generalisation matrices were averaged across patients.

2.3.3.2. Experimental questions

2.3.3.2.1. Relative decoding accuracy using spectral frequency power or phase vs. voltage

We first assessed whether power, phase and voltage each contained sufficient information to enable decoding. We therefore created both voltage feature vectors and frequency feature vectors of power and phase data (separately) using all 60 frequencies between 4 and 200 Hz and used these as input to classifiers. We compared each group-average timecourse to chance (0.5) using one-tailed, one-sample t-tests. We compared power and voltage, and phase and voltage, using paired t-tests. Probabilities were adjusted to control the false-discovery rate at $\alpha = 0.05$ (Benjamini & Hochberg, 1995). We confirmed that near-identical patterns of results were obtained in the 8 patients with left-hemisphere electrodes previously analysed by Rogers et al. (2021), the 7 patients with left-hemisphere

electrodes analysed for the first time in this work, and the 3 patients with right-hemisphere electrodes (Supplementary Materials S1, Figure S5). Having done so, we combined all patients into a single group for all subsequent analyses.

We next investigated whether there were differences in decoding performance between power and phase within different frequency ranges. We therefore divided the 60 frequencies into theta (4 – 7 Hz, 11 frequencies), alpha (8 - 12 Hz, 7 frequencies), beta (13 – 30 Hz, 13 frequencies), gamma (30 – 60 Hz, 10 frequencies) and high gamma (60 – 200 Hz, 19 frequencies) ranges. We constructed frequency vectors of power and phase data using only power or phase at frequencies within each range and used these as input to separate classifiers. We compared each group-average timecourse to chance and then compared each range to the timecourse of decoding using all 60 frequencies.

2.3.3.2.2. Does the time-frequency semantic code change dynamically?

We then tested whether, if either power or phase data contained sufficient information to enable decoding, the power or phase code exhibited the same distributed, dynamic properties identified by Rogers et al. (2021) in both the voltage data (for the first subset of patients) and the computational hub-and-spoke model. We tested for these properties using both frequency vectors constructed using all 60 frequencies and vectors constructed using only frequencies within each range. Constant decodability had already been assessed by one-sample t-tests of each group-average timecourse against chance.

To assess local temporal generalisation we inspected the group-average generalisation matrices generated by testing each classifier at every possible timepoint. To test whether the patterns observed were statistically significant, we identified the best-performing classifier at each timepoint and then conducted paired t-tests to assess whether classifiers trained at every other possible timepoint performed equally well. Probabilities were adjusted to control the false-discovery rate at $\alpha = 0.05$ (Benjamini & Hochberg, 1995).

To assess the shape of the generalisation window over time we computed the pairwise cosine distance between each row of the generalisation matrix, then applied agglomerative hierarchical clustering using *hclust* (native R) and cut the tree to create 10 clusters. We selected this cluster number to make these results comparable to those of Rogers et al. (2021). We averaged the timecourses of classifiers within each cluster and

inspected performance over time (again, probabilities were adjusted to control the false-discovery rate). To quantify the patterns observed we took classifiers trained every 50 ms (i.e., 0 ms, 50 ms, 100 ms ...; this ensured that, for classifiers trained on voltage, consecutive 50 ms windows did not overlap and were therefore independent). We calculated the area under the curve between each timecourse and a horizontal line at chance (0.5) and fitted a piecewise linear regression to the area values using the *segmented* library in R, using *segmented* with the Bayesian Information Criterion to determine the number and location of breakpoints to produce the best model given the number of free parameters.

To assess and visualise changes in code direction we first took the classifiers trained on all data at each timepoint and then (separately for every electrode, every patient and every timepoint) calculated the mean classifier coefficient within each frequency range (or over all 60 frequencies, or over the 50 ms within each window in the case of voltage). These mean classifier coefficients were converted to one NIFTI volume per patient and per timepoint, at 2.5 mm isotropic resolution in MNI space (MNI152NLin2009cAsym), using SPM12 (<https://www.fil.ion.ucl.ac.uk/spm/>) implemented in MATLAB r2023b. One patient (Patient 20) was excluded from this analysis because no MNI coordinates were available for their electrodes. Other patients were missing MNI coordinates from only a subset of electrodes; we included those patients in the analysis but ignored coefficients on the electrodes with missing coordinates. We projected coefficients in each volume to the pial surface (fsaverage template; (Dale et al., 1999; Fischl et al., 1999) using *-volume-to-surface-mapping* with trilinear interpolation and smoothed them on the surface at 6 mm FWHM using *-metric-smoothing* in Connectome Workbench 1.5.0. We calculated the proportion of negative coefficients that each vertex received (since animals were coded as 0 and inanimate objects as 1, a negative coefficient indicates that an increase in power is associated with increased probability that the stimulus is animate). We plotted these proportions on the pial surface using *plot_surf_stat_map* in *nilearn* 0.10.3 implemented in Python 3.9. Finally, we animated the results – ffmpeg 5.1.6 (<https://ffmpeg.org/>) was used to concatenate plots of coefficients at successive timepoints with a frame rate of 10 (10 times slower than real time) and motion interpolation.

3. Results

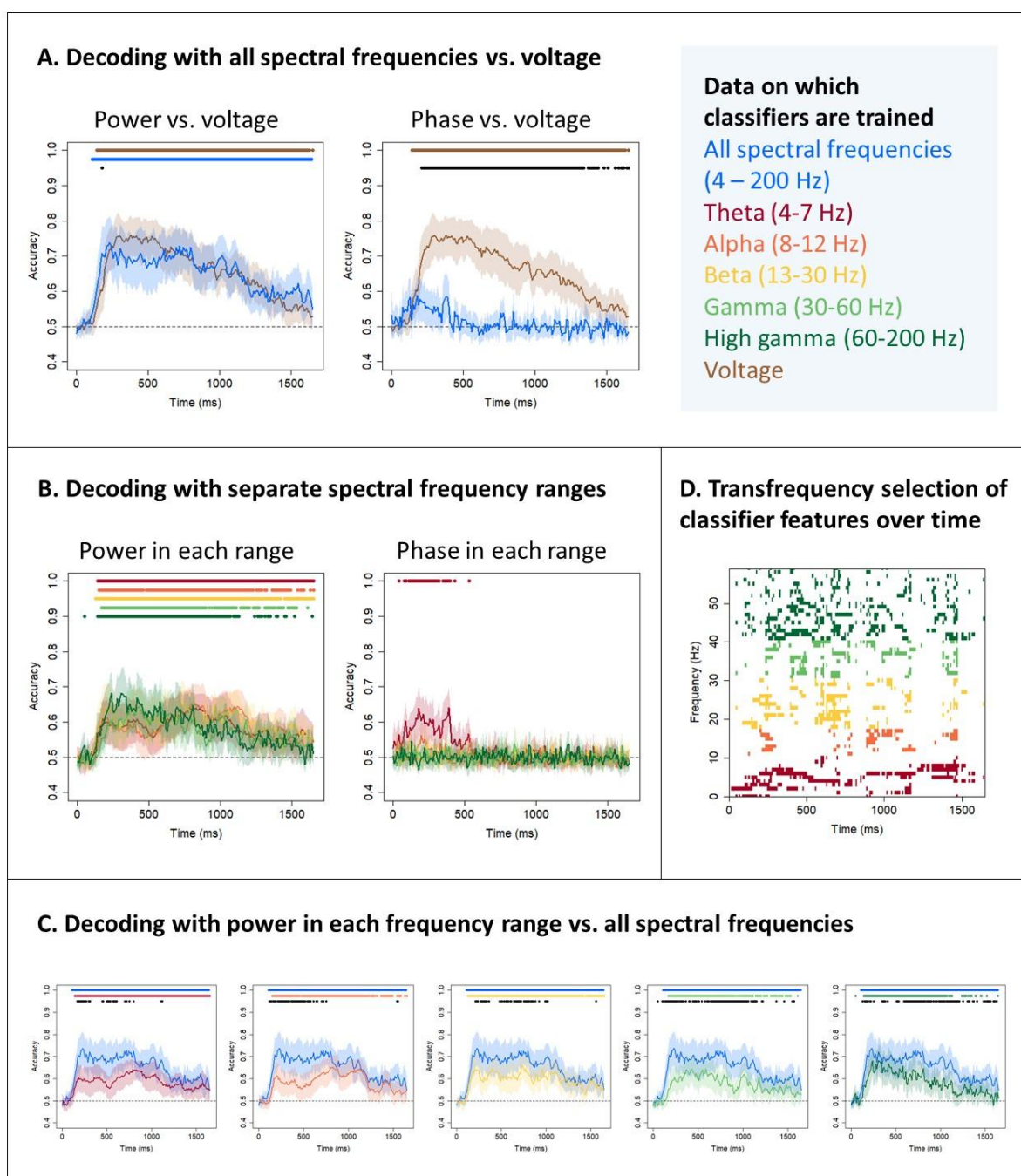
3.1. Relative decoding accuracy using spectral frequency power or phase vs. voltage

We first tested whether it was possible to decode the animacy of the stimuli using time-frequency power and/or phase and whether decoding performance was comparable to voltage (Rogers et al., 2021). We compared classifiers trained on frequency features of power or phase at 60 frequencies, logarithmically spaced between 4 and 200 Hz, to classifiers trained on voltage features. Figure 2A shows the results. Time-frequency power showed an almost identical decoding profile to voltage – hold-out accuracy rose to around 0.7 at 200 ms post stimulus onset and remained significantly above chance (0.5) throughout the time window. At no point was it possible to decode animacy from classifiers trained on phase.

Next, to disentangle the role of local neural activity versus distal input, we trained classifiers on frequency features composed only of power or phase values from one frequency range - theta (4 – 7 Hz), alpha (8 – 12 Hz), beta (13 – 30 Hz), gamma (30 – 60 Hz) and high gamma (60 – 200 Hz). Figure 2B shows the decoding profile in each range. For power, the patterns were almost indistinguishable, showing the same rise to about 0.6 and remaining above chance for most of the time window of interest. On the assumption that gamma and high gamma frequencies represent local neural activity, these results confirmed that local neural activity in the vATL expresses semantic information. However, we found that power in every range was sufficient, and no range was necessary, for above-chance decoding. For phase, decoding was rarely above chance, with the exception of weak decoding using theta data in the window 0 - 500 ms. The remaining analyses therefore focused on characterising semantic information encoded in time-frequency power.

Figure 2C shows hold-out accuracy for classifiers trained on power within a single range compared to accuracy for classifiers trained on power at all 60 frequencies. For each range there was a time window during which decoding on all frequencies reached its peak and decoding on a single range performed significantly less well. This indicated that no individual range contains sufficient information to enable decoding accuracy comparable to accuracy based on all frequencies. To investigate which combination of frequencies enabled such successful decoding, we inspected the coefficients on each frequency. Figure 2D shows

461 these averaged across electrodes for one example patient (similar plots for all patients are
462 shown in Supplementary Materials S1, Figure S1; plots showing coefficients on individual
463 electrodes for the one sample patient are shown in Supplementary Materials S1, Figure S2).
464 Throughout the time window, frequencies across the whole range were selected by the
465 classifier and thus contributed to decoding performance. This is evidence that the vATL
466 represents semantic information via a *transfrequency* code, by which we mean that
467 frequencies within the theta, alpha, beta, gamma and high gamma ranges all contribute to
468 information representation.



469

470 Figure 2: Semantic decoding with time-frequency power and phase. (A) Mean and 95%
471 confidence interval of the hold-out accuracy for classifiers trained on power or phase
472 frequency features for all frequencies between 4 and 200 Hz (blue) or on voltage features
473 (brown). Coloured dots indicate a significant difference between classifier accuracy and
474 chance (0.5, one-sample t-test with probabilities adjusted to control the false-discovery rate
475 at $\alpha = 0.05$). Black dots indicate a significant difference between accuracies at a given
476 timepoint (paired t-tests with probabilities adjusted to control the false-discovery rate at $\alpha =$
477 0.05). (B) Mean and 95% confidence interval of the hold-out accuracy for classifiers trained
478 on frequency features composed only of power or phase values from a given range - theta
479 (4 – 7 Hz, red), alpha (8 – 12 Hz, orange), beta (13 – 30 Hz, yellow), gamma (30 – 60 Hz, light
480 green) and high gamma (60 – 200 Hz, dark green). Coloured dots indicate a significant

difference between classifier accuracy and chance (0.5, one-sample t-test with probabilities adjusted to control the false-discovery rate at $\alpha = 0.05$). (C) Mean and 95% confidence interval of the hold-out accuracy for classifiers trained on frequency features composed only of power or phase values from a given range compared to classifiers trained on frequency features including all frequencies between 4 and 200 Hz. Coloured dots indicate a significant difference between classifier accuracy and chance; black dots indicate a significant difference between accuracies. (D) Feature selection of each frequency at each timepoint (averaged across electrodes) for classifiers trained on frequency feature vectors including all frequencies between 4 and 200 Hz for a single patient. Features with a nonzero coefficient are shown in colour.

3.2. Does the time-frequency semantic code change dynamically?

Our second question was whether the time-frequency code changes dynamically over stimulus processing in a manner similar to the voltage code. The preceding results show that time-frequency power exhibits one property also observed in voltage, namely constant decodability - animacy can be decoded from whole-spectrum power, and from power within each frequency range independently, from about 100-200ms post onset until about the mean onset of naming (1190ms in the subset of patients analysed by Rogers et al., 2021). Next, we applied the method of temporal generalisation to investigate whether time-frequency power, like voltage, shows (a) local temporal generalisation, indicating a changing neural code, and (b) a widening generalisation window, indicating rapid initial change that slows over time (Rogers et al., 2021).

Figure 3A shows the generalisation profile of classifiers trained on power at all frequencies (left) and those trained on voltage (right). The diagonal of each matrix indicates the hold-out accuracy of classifiers trained and tested at the same timepoint. Points closer to the diagonal indicate how well a classifier generalises to neighbouring timepoints, while points far from the diagonal indicate how well it generalises to timepoints further away. Local temporal generalisation is signified by better decoding close to the diagonal. In these plots, saturated colours indicate classifier accuracies statistically indistinguishable from the best-performing classifier at the same timepoint. Desaturated colours indicate accuracies that are above chance, but that are reliably worse than the best-performing classifier. Grey indicates classifiers that did not exceed chance performance.

The superimposed squares on each plot provide illustrative referents for evaluating whether the temporal generalisation pattern signifies a dynamically changing code. If

classification accuracy is significantly higher for the two on-diagonal corners than the two off-diagonal corners, this signifies that (a) the neural signal reliably encodes semantic information at the two timepoints but (b) the signal changes in nature – whatever features of the signal support the good performance at the training timepoint are not present at the more remote testing points. This pattern is clearly observed for both voltage and whole-spectrum power (Figure 3A), indicating that the transfrequency semantic code exhibits dynamic properties similar to voltage.

When decoding from each range separately, however, the local generalisation pattern was less clear overall, and arguably present only in lower frequency ranges (theta, alpha, and beta; Figure 3B). Gamma and high gamma showed a different pattern: classifiers fitted at a given timepoint generalised approximately equally well to all prior timepoints back to the first timepoint where decoding was reliable (about 100-200ms post stimulus onset). That is, classifiers trained on gamma and high gamma showed excellent *retrograde* generalisation. This pattern yields the vertical “edge” clearly evident in the high gamma matrix, and blurrier but still present in the gamma matrix. The classifiers also generalised to future timepoints, but with performance often tailing off for more remote future timepoints — that is, they showed diminishing *anterograde* generalisation.

Together these observations suggest two interesting things about the semantic code as expressed within and across different frequency ranges. First, whatever information a classifier trained on gamma or high gamma power exploits at its own training timepoint must also be present in the signal at all earlier time points, all the way back to the first moment semantic information appears (producing good retrograde generalisation). However, information exploited by a classifier at its training timepoint may *not* be present in the signal at future timepoints, suggesting that these features “drop out” over the course of stimulus processing, producing a graded drop-off in anterograde generalisation. Second, the dynamically-changing pattern observed in the transfrequency code may be carried predominantly by lower frequency information.

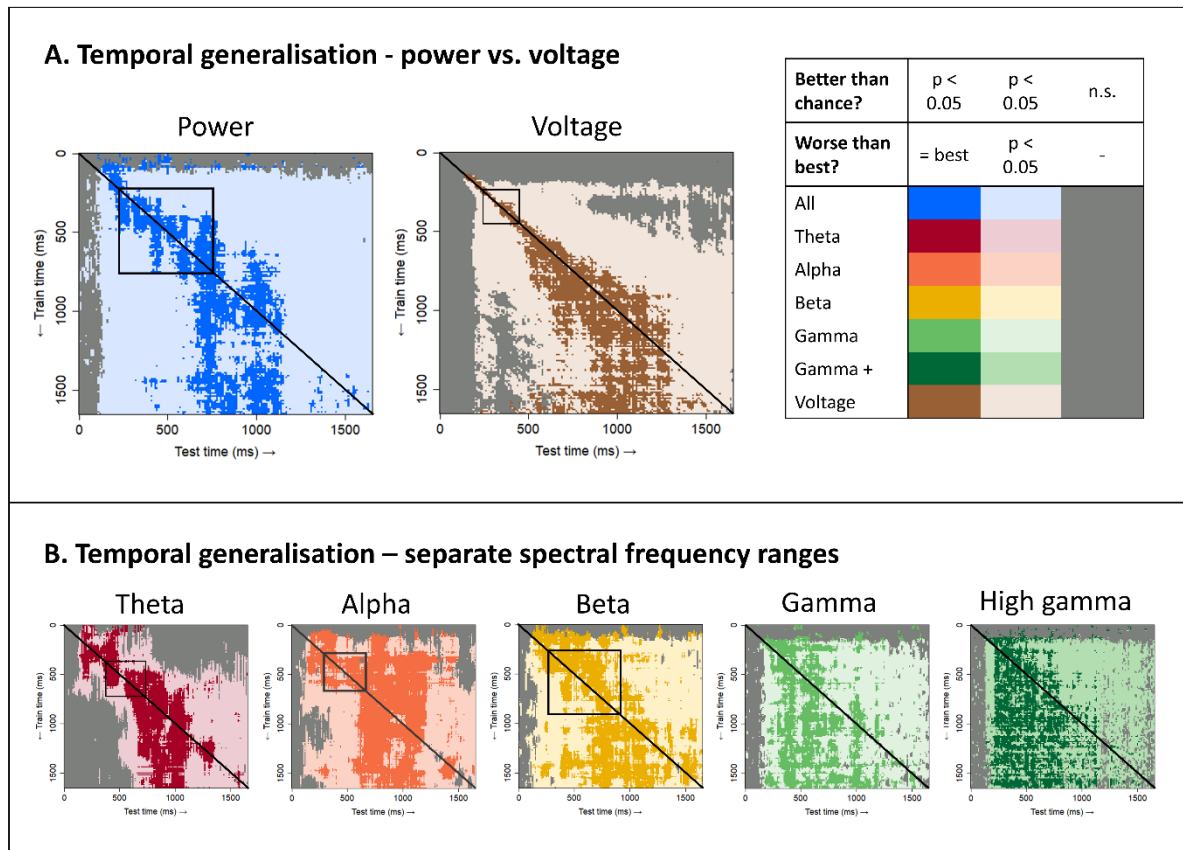


Figure 3: Temporal generalisation of classifiers. For each test time, the train time (vertical dimension) of the best classifier, plus the train times of other classifiers that perform statistically indistinguishably from the best ($p > 0.05$, paired t-test with probabilities adjusted to control the false-discovery rate at $\alpha = 0.05$), are shown in the most saturated colour. Desaturated colours show the train time of classifiers exhibiting a significant difference from the best classifier ($p < 0.05$), but which still perform significantly better than chance (0.5, one-sample t-test with probabilities adjusted to control the false-discovery rate at $\alpha = 0.05$). Timepoints where classifiers trained at that timepoint do not perform significantly better than chance are shown in grey. Local temporal generalisation is indicated by significantly better performance on timepoints at or near the train time (the diagonal shown on each plot) than at more remote timepoints. The squares highlight cases where hold-out accuracies at two training timepoints (the two corners on the diagonal) are both significantly better than when the same classifiers are used to decode more remote timepoints (the two off-diagonal corners). (A) Results for classifiers trained on power frequency features for all frequencies between 4 and 200 Hz (blue) or on voltage features (brown). (B) Results for classifiers trained on power frequency features from a single range - theta (4 – 7 Hz, red), alpha (8 – 12 Hz, orange), beta (13 – 30 Hz, yellow), gamma (30 – 60 Hz, light green) and high gamma (60 – 200 Hz, dark green).

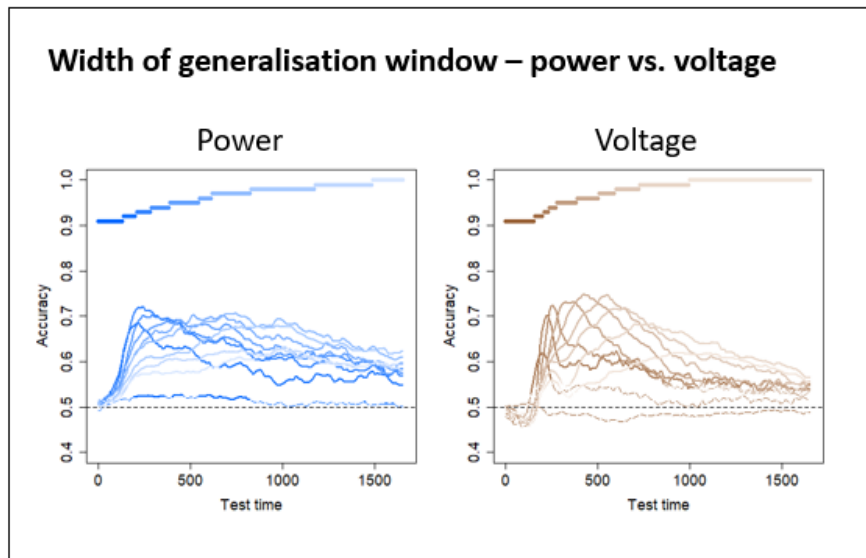


Figure 4: Width of generalisation window. Classifiers trained on power frequency features for all frequencies between 4 and 200 Hz (shades of blue) or on voltage features (shades of brown) were grouped into ten clusters via agglomerative hierarchical clustering (see Methods). The coloured bars indicate which training timepoints belong to each cluster. The “timecourses” show the mean hold-out accuracy for classifiers within each cluster at each timepoint. Lines are solid where there was a significant difference between classifier accuracy and chance (0.5, one-sample t-test with probabilities adjusted to control the false-discovery rate at $\alpha = 0.05$) and dashed where there is no significant difference. Coloured bars show the grouped timepoints in each cluster. Statistics supporting this pattern are reported in Supplementary Materials S1, Figure S4).

We decided to focus subsequent analyses on frequency ranges in which a dynamic code was most clearly observed, namely voltage and whole-spectrum power. We next assessed whether the window of above-chance temporal generalisation widened over time, consistent with an overall slowing of representational change. To this end, we grouped the classifiers into ten clusters based on similarity in their temporal decoding profiles (see Methods). That is, classifiers that show similar patterns of prediction accuracy across all test timepoints were clustered together, and we computed the mean accuracy profile over time for each cluster. Figure 4 shows the result. The coloured bars (above) indicate the which training timepoints were grouped by the clustering analysis and the “timecourses” (below) illustrate the mean accuracy hold-out accuracy for classifiers with that cluster at each timepoint. Classifiers trained earlier showed a sharper rise and fall in accuracy, whereas classifiers trained later exhibited a wider window of generalisation (similar plots for classifiers trained on separate frequency ranges are shown in Supplementary Materials S1,

Figure S3; statistics supporting this claim are reported in Supplementary Materials S1, Figure S4). Thus both voltage and transfrequency power show a slowing of dynamic representational change over the course of stimulus processing.

Finally, we assessed whether the observed dynamic change was accompanied by change in the apparent code direction—that is, whether an increase in voltage or in power predicted a greater likelihood that the stimulus is an animal at some points in time, and a greater likelihood that it is inanimate at others. Change in code direction is another hallmark of nonlinear representational change (Rogers et al., 2021). To see this, we aggregated the classifier coefficients across patients for each timepoint and plotted these on the cortical surface. Specifically, at each cortical location, we first calculated the number of electrodes across patients that had a non-zero coefficient, then computed the proportion of these having a negative coefficient (since animals were coded as 0 and inanimate objects as 1, a negative coefficient means that an increase in voltage or power is associated with increased probability that the stimulus is an animal). Figure 5 shows these proportions at 4 different timepoints post stimulus onset, for classifiers trained on whole-spectrum power (top) or voltage (bottom). Warm colours indicate areas and timepoints where most selected electrodes have negative coefficients, while cool colours indicate areas and timepoints where most selected electrodes have positive coefficients. Green shades indicate areas and timepoints where selected electrodes have a mixture of positive and negative coefficients. We also animated the coefficient change over time, with videos for classifiers trained on all frequencies, individual frequency ranges, and voltage available at https://github.com/slfrisby/ECoG_LASSO/.

For both voltage and power, changes in code direction were observed. For voltage, for instance, the anterior-most part of the ventral temporal lobe was uniformly positive at 200ms and uniformly negative at 600ms. By 1000ms, midway between the anterior and posterior extent of coverage, coefficients showed a medial-to-lateral change in direction, suggesting that voltage increases signify animals in the more medial portion and non-animals in the more lateral portion. Whole-spectrum power also showed changes in code direction over time; for instance, in Figure 5, the medial anterior portion of ventral temporal cortex showed a negative relationship between power and animacy at 400ms, but a positive relationship by 1000ms. At 1000ms, the transfrequency coefficients also showed a medial-

to-lateral change, with increased power signifying animals all along the medial extent of the coverage area, but showing a more mixed pattern across the more lateral extent.

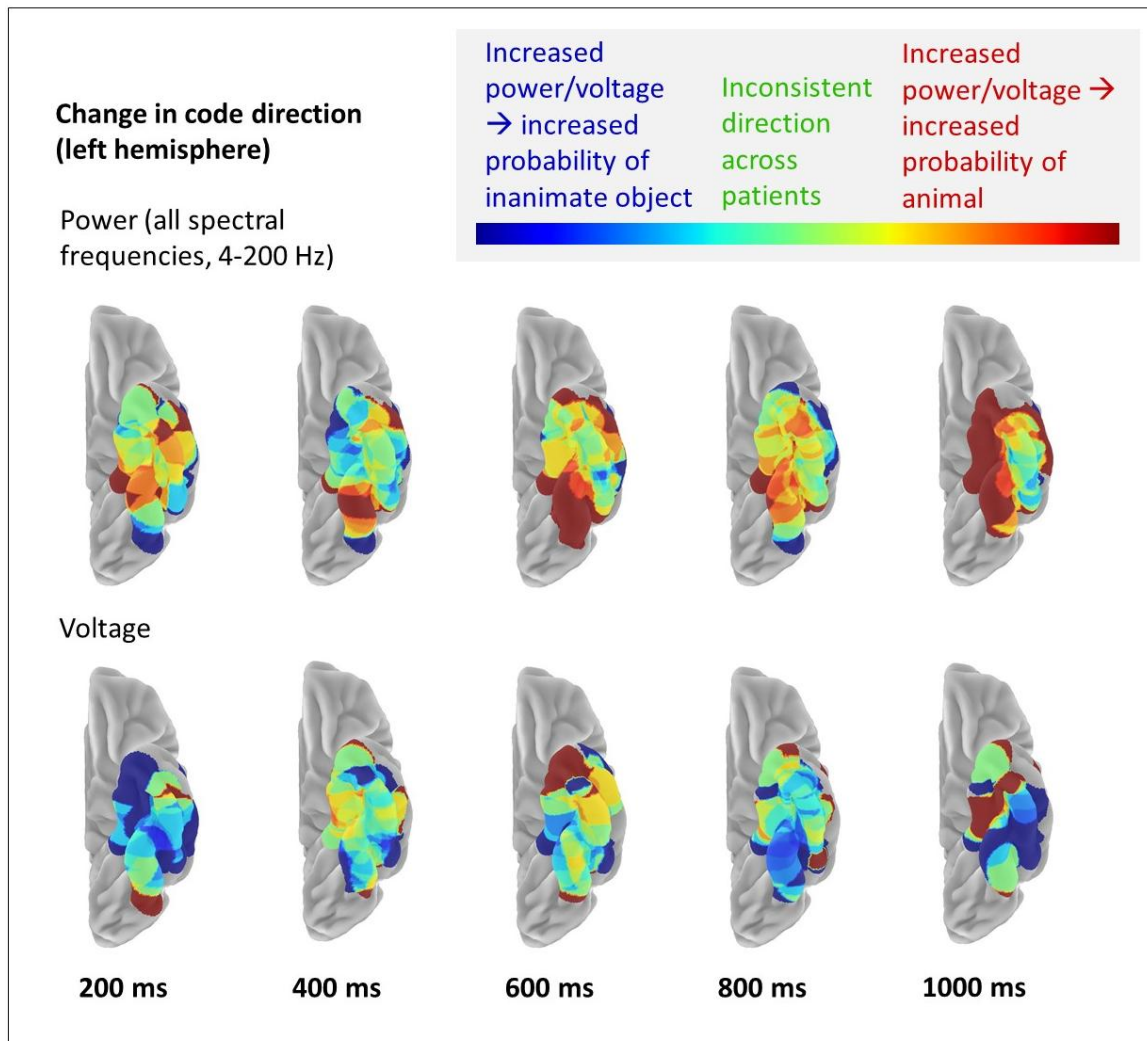


Figure 5: Change in code direction. Coefficients for classifiers trained on power frequency features for all frequencies between 4 and 200 Hz, or on voltage features, are projected onto the ventral surface of the left hemisphere. Time is shown relative to stimulus onset. Colours indicate the proportion of classifier coefficients at that location with the same sign (see Methods). Warm colours indicate that coefficients are mostly negative across patients (since animals were coded as 0 and inanimate objects as 1, a negative coefficient indicates that an increase in power is associated with increased probability that the stimulus is animate); cool colours indicate that coefficients are mostly positive across patients (an increase in power is associated with increased probability that the stimulus is inanimate); green shades indicate that the region was selected in multiple patients but that the sign of the coefficient is not consistent across patients.

4. Discussion

While it has long been clear that the vATLs play an important role in human semantic cognition (Lambon Ralph et al., 2017; Patterson et al., 2007; Rogers et al., 2004), precisely *how* neural populations encode such information remains uncertain. Recent studies applying multivariate decoding to voltages recorded from the cortical surface suggest that semantic information is expressed in a rapidly-changing code (Rogers et al., 2021), but, with respect to uncovering local neural representations, such studies remain limited, since voltages can be influenced by a variety of sources both local and remote. The current work instead applied multivariate decoding to a time-frequency decomposition of ECoG data, with two results that advance our understanding of the vATL semantic code in important ways. First, information about animacy is expressed by power in all frequencies simultaneously - it appears within each range considered independently, and jointly across all ranges, for most of the duration of stimulus processing. Moreover, decoding accuracy was reliably better when models were fitted to all frequencies simultaneously compared to each individual range; and classifiers fitted to the whole spectrum always selected features from all frequency ranges no matter which timepoint they were trained on. This result strongly suggests conjoint encoding of semantic information across the power spectrum (Frisby et al., 2023). Second, decoding from both power and voltage revealed characteristics of dynamic representational change: constant decodability, local temporal generalisation, a widening generalisation window, and changes in code direction. This is likely to be driven by low frequencies – gamma and high gamma classifiers showed a contrasting pattern of excellent retrograde generalisation back to the earliest successful decoding timepoint, but anterograde generalisation that diminished for more remote timepoints.

These findings agree with multiple univariate studies that variously associate theta (Hermes et al., 2014), alpha (Clarke et al., 2018), beta (Abel et al., 2015; Sato et al., 2021), gamma and high gamma (Arya, 2019; Bartoli et al., 2019; Cervenka et al., 2013; Chan et al., 2011; Crone et al., 2001; Crone & Hao, 2002; Edwards et al., 2010; Forseth et al., 2018; Kojima et al., 2013; Nakai et al., 2017, 2019; Snyder et al., 2023; Tanji, 2005; Wang et al., 2011) with semantic tasks. One interpretation of our results is that information is represented redundantly in multiple frequencies. A second possibility is that, since the brain exhibits properties of a dynamical system (Jones et al., 2015; Yuste, 2015), rhythmic patterns may be an epiphenomenon of normal functioning or even of epileptiform activity

(Drane et al., 2008; Hamberger, 2007; Henry et al., 1998). Therefore, some frequency ranges may represent information not in a “strong” sense (meaning that the brain uses time-frequency power to represent information for downstream computation) but in a “weak” sense (meaning that time-frequency power is decodable but is not used as the principal means of representation; Watrous et al., 2015).

However, the finding that classifiers trained on whole-spectrum power outperform those trained on a single frequency range and select features from across the frequency spectrum (Figures 2C and 2D) supports a more nuanced conclusion. Classifiers trained with LASSO regularisation prefer sparse solutions, in which only a small subset of features receive nonzero coefficients. If the information in the different frequency ranges were redundant, a classifier would not assign nonzero coefficients to many features that spanned the frequency range. Instead, these results support the notion of *transfrequency* representation – information is encoded *across* multiple frequency ranges (sometimes called frequency “multiplexing”; (Murphy, 2025; Watrous et al., 2015). Some previous univariate studies have proposed specific roles that different frequency ranges may play – abstraction from multiple episodic experiences (Sauseng & Klimesch, 2008), binding multiple perceptual or semantic features of an item (Merker, 2013; Sauseng & Klimesch, 2008; Stryker, 1989; Tallon-Baudry & Bertrand, 1999; von der Malsburg, 1995), categorisation (Bartoli et al., 2019), encoding semantic similarity (Solomon et al., 2019; Spiers, 2020), and enabling retrieval independent of sensory modality (Sauseng & Klimesch, 2008) in a way that is appropriate for the context (Halgren et al., 2015; Jackson, 2021; Klimesch, 2012; Marko et al., 2019). Successful decoding of animacy is a finding that is consistent with all of these possible functions - to test these possibilities, future studies should employ methods capable of decoding fine-grained information (Y. Chen et al., 2016; Clarke, 2020; Cox et al., 2024; Rupp et al., 2017).

Rather than positing different cognitive functions for different frequency bands, a parsimonious assumption is that the frequency of activity is related to varying temporal dynamics across neural circuits. A much rehearsed notion is that the frequency of (at least some) activity is inversely related to transmission distance (Canolty & Knight, 2010; Kopell et al., 2000; Lachaux et al., 2012). On this assumption, the notion of transfrequency representation is highly consistent with the short and long-range connections in the hub-and-spoke model (Lambon Ralph et al., 2017; Patterson et al., 2007; Rogers et al., 2004).

699 According to this model, neither the vATL “hub” nor the modality-specific “spokes” are
700 individually sufficient for full semantic cognition – rather, information is encoded conjointly
701 in multiple regions and in the longer-range interactions between them (Frisby et al., 2023).
702 We note here that the assumption that frequency is inversely related to distance is known
703 to be violated under some circumstances. Individual neurons cannot fire at high gamma
704 frequencies, meaning that high gamma activity almost certainly reflects the asynchronous
705 activity of local populations, but it is possible for low-frequency activity to occur in the
706 absence of any long-range connections (Florez et al., 2015).

707 Overall, future studies should seek to test the hub-and-spoke model’s claims more
708 directly and to adjudicate between the hub-and-spoke model and competing theories
709 (Binder & Desai, 2011; Clarke et al., 2018; A. R. Damasio, 1989; H. Damasio et al., 2004;
710 Fauconnier & Turner, 1998; Frisby et al., 2025; Huth et al., 2016; Martin, 2007, 2016; Tranel
711 et al., 1997; Tyler et al., 2000). The hub-and-spoke model’s claims include not only (1) that
712 information is encoded via interactions between hub and spokes but also, in contrast to
713 other theories, (2) that the hub is bilateral and (3) that the hub is centred on a specific
714 region of vATL located within the larger swathe analysed here (Lambon Ralph et al., 2017;
715 Shimotake et al., 2015); but see also (Rogers et al., 2021). Testing the first of these claims
716 will require electrode coverage not only of the vATL but also of spoke regions; testing the
717 second will require a larger sample size of right hemisphere patients than is reported here
718 ($n = 3$ compared to $n = 15$ in the left hemisphere).

719 We were unsuccessful in decoding semantic information from time-frequency
720 phase in any frequency range, apart from a short period of significant decoding using phase
721 in the theta range between 0 and 500 ms post stimulus onset. It is possible that phase was
722 variable across trials, and so averaging data across trials before extracting phase resulted in
723 data that no longer correlated with animacy. However, even if this result is a false negative,
724 any successful decoding with time-frequency phase poses questions of interpretability. One
725 possibility is that phase “weakly” represents information (Watrous et al., 2015) – for
726 example, animacy may be decodable from theta phase because gamma amplitude is locked
727 to the phase of theta activity (Aru et al., 2015; Benítez-Burraco & Murphy, 2019; Canolty et
728 al., 2006; Canolty & Knight, 2010; Hermes et al., 2014; Heusser et al., 2016; Lisman &
729 Jensen, 2013; Sederberg et al., 2003). Decoding using phase-amplitude coupling metrics as

input features has been successful within the field of speech production (Proix et al., 2022) and future research could apply a similar method to clarify the contribution of phase-amplitude coupling to semantic representation. An alternative possibility is that theta phase really does represent semantic information in a “strong” sense (Watrous et al., 2015), although the mechanism via which a finite set of phase values might represent the complex semantic properties of a near-infinite number of concepts is not intuitively obvious.

This study was the first to investigate whether activity in different frequency ranges exhibits the same pattern of dynamic representational change seen in voltage (Rogers et al., 2021). Dynamic change was clearly evident when decoding whole-spectrum power, just as it was observed in a recurrent deep neural network model of semantic processing. It is known that the individual units of deep layers of a computational model do not readily correspond to interpretable features – yet the activity pattern *across* units is more similar for more similar concepts, and differentiation between concepts becomes progressively more fine-grained over time. For a detailed theoretical of this *distributed, dynamic* code, how it relates to observed changes in coefficients, and alternative kinds of code, we refer the reader to Rogers et al. (2021). However, gamma and high gamma showed a different pattern – excellent retrograde generalisation and diminishing anterograde generalisation. The good retrograde generalisation indicates that features exploited by a classifier at its training timepoint must also be present, and must signify the same information in the same way, for all preceding timepoints (up to the earliest successful timepoint). That is, this part of the code must be relatively *static* over processing from the earliest arrival of perceptual signals up to the time where the classifier is fitted. Conversely, the diminishing anterograde generalisation indicates that features exploited by a classifier at its training timepoint eventually fall away at future timepoints.

Figure 6 illustrates in schematic form what such a decoding pattern might suggest about how animacy is encoded in high-frequency activity (and thus, by inference, in local neural activity). Each curve represents a hypothetical response to animate stimuli in the gamma or high gamma range for one electrode over time. Every electrode shows increased gamma power for animate stimuli from early in processing, but the timecourse of this change differs between electrodes. Where gamma increases most strongly early in processing, it also diminishes most rapidly; where gamma increases are smaller initially,

they persist longer. Since the LASSO penalty prefers to assign features the smallest coefficients possible, models fitted with LASSO regularisation will select whichever electrodes show the strongest response at the training timepoint. Thus models fitted to the earliest timepoints will select the red electrodes, those fitted slightly later will select the orange ones, etc. Generalisation to future timepoints will diminish as the gamma power decreases on the selected electrodes, producing diminishing anterograde generalisation. Because all electrodes carry useful signal from the onset, however, each classifier generalises successfully to all prior time points, producing good retrograde generalisation. The generalisation pattern is thus consistent with the view that animacy is encoded in the local activity of many neural populations, some responding strongly but then quickly diminishing, and other initially responding less strongly but persisting for a longer duration.

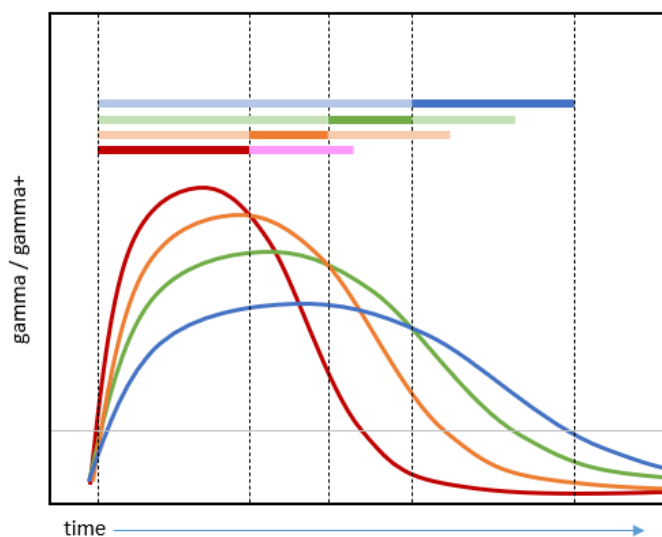


Figure 6: Hypothesised semantic code in gamma and/or high gamma producing full retrograde generalisation and limited anterograde generalisation. Each curve represents the hypothesised gamma/high gamma response to animate stimuli by one electrode. Where curves are above the grey horizontal line, they encode information about animacy. All electrodes encode animacy from early in processing, but models regularised with the LASSO penalty will place coefficients on whichever electrode has the strongest response at a given timepoint. The saturated parts of the coloured horizontal bars thus indicate which electrode is selected by a classifier at each timepoint. The desaturated parts indicate the generalisation profile of the resulting classifier. Electrodes that show the strongest responses early on diminish in power most rapidly, limiting anterograde generalisation. Because all electrodes carry information about animacy from the onset of the signal, however, all classifiers generalise well to the beginning of the processing window.

Finally, the current work carries important implications for interpreting other sources of information about semantic representation. Noninvasive methods such as EEG and MEG cannot discern activity within high gamma frequencies because it is blocked by the skull and the scalp. The current work suggests that dynamic representational change is mainly observed in lower frequency ranges that are more readily measurable via these other techniques—thus the comparatively static code revealed in high frequency ranges may be obscured in other imaging modalities. This highlights the utility of intracranial techniques for revealing information representation, both in humans and in non-human primates (although note that primate and human studies can vary in their conclusions – for example, this study finds that animacy is represented more anteriorly than previous work found in macaques; Bao et al., 2020). Additionally, the current work highlights circumstances in which representations in neural activity differ in their nature from representations observed in neuro-computational models (which exhibit the dynamic properties observed in voltage and transfrequency power, not the comparatively static code observed in high frequencies). Further work should explore the circumstances under which representational properties in brains and machines converge and diverge (Cox et al., 2015).

5. Conclusion

To summarise, we have shown that it is possible to decode semantic information from every frequency range between theta and high gamma. Although individual frequency ranges contained enough information to drive decoding above chance, decoding from the whole frequency spectrum at once produced the highest decoding accuracy, equivalent to using voltage. Decoding from the whole frequency spectrum also revealed the same set of dynamic properties observed in voltage; however decoding of high-frequency ranges alone showed a qualitatively different pattern, suggesting that local firing activity expresses a more static semantic code. Together these results provide an important step toward understanding how neural systems encode semantic information.

Data and code availability: We are unable to share raw data for this study because the patients did not provide informed consent to do so. However, matrices containing power,

phase, and voltage features (columns) for each stimulus (rows) will be made available upon peer review and acceptance. Code is available at https://github.com/slfrisby/ECOG_LASSO/.

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